

3D quantum Hall effect was successfully observed in recent ZrTe_5 and HfTe_5 experiments under strong magnetic fields. While the initially proposed CDW was not demonstrated in subsequent experiments, the underlying mechanisms are still under debate. The authors of this manuscript (MS) consider the electrons coupled to the boson fields in 3D quantum Hall systems and find that without CDW, the system can exhibit inevitable first-order phase transitions when the number of occupied LLs jumps. Then the system may be driven into a phase separation state with percolation transitions. Thus, the experimental results, including the quasi-plateaus Hall resistivity and the metal-insulator transition, can be explained. This MS adopts a dimensionless model, which greatly simplifies the calculations and has certain universality.

I suggest that the authors address the following concerns.

1. The authors use both quadratic dispersion and massive Dirac dispersion to describe the low carrier density metals ZrTe_5 and HfTe_5 . In the MS, many calculations and arguments are only based on quadratic dispersion, such as Eq. (5) and below and Fig. 3. Moreover, the quasi-plateaus Hall resistivity and the metal-insulator transition are also derived from quadratic dispersion. My question is whether the simplified quadratic dispersion model can correctly capture the low-energy behaviors in ZrTe_5 and HfTe_5 , since their ground states were suggested to lie near the phase boundary between the strong topological insulator and weak topological insulator?

2. The authors assume that the mean-field components $\Delta_{\pm 2k_F}$ are ignored and only Δ_0 is considered, which are key points of their calculations. To my understanding, whether Δ_0 or $\Delta_{\pm 2k_F}$ dominates depends on the lowering of the free energy F , but not their relative magnitudes. Please give more evidences of the F variations due to Δ_0 and $\Delta_{\pm 2k_F}$. On the other hand, the common CDW theory only focus on the wave vector $Q = \pm 2k_F$. Which physical factors determine the dominance of $Q = 0$ in this MS?

3. The authors calculate the mean field components Δ_0 and Δ_{2k_F} , and the results are shown in Fig. 1(d), in which the parameters are set as $\mu = 1$ and $B = 2$. What values are the Fermi momentum $k_{F,n}$? It seems that $k_{F,n}$ is determined by $\mu = \epsilon_{k_{F,n}} = \frac{1}{2}k_F^2 + B(n + \frac{1}{2})$, from which $n = 0$ and $k_F = 0$? Please check it.

4. In Fig. 2(c), do the black lines represent the renormalized LLs? Without the boson couplings, the LLs are simply $\epsilon_{0,n}(B) = B(n + \frac{1}{2})$; with the boson couplings, the LLs

are renormalized as $\epsilon_{0,n}(B) - D\Delta$? What values do the mean field Δ and the fixed chemical potential μ take for each LL in the calculations?

5. In the third paragraph of Page 1, the authors write “Increasing the disorders could weaken and eventually eliminate the first order transitions”. However, the role of disorder is only briefly discussed in the last section, which cannot be regarded as the main results and summarized in the third paragraph of Page 1.

6. In the paragraph below Eq. (4) of the definition of Δ_N , should $\epsilon_{N-1,0}(B)$ in the numerator should be revised as $\epsilon_{0,N}(B)$? Please check it.